Amrit Gopinath

Third-Year B.E. CSE Undergraduate • NLP, LLM Interpretability, and Multimodal Research

amrit2410182@ssn.edu.in | github.com/Amrit828 | LinkedIn | Portfolio

Education

Sri Sivasubramaniya Nadar College of Engineering, Chennai 2024–Present

B.E. Computer Science and Engineering | CGPA: **9.383/10.0** | Department Rank: **1st**

Research Focus

Multilingual NLP and representation analysis; language-model interpretability and model introspection; mixture-of-experts routing; growing research interest in vision–language models and AI safety.

Selected Research

A Declarative–Procedural Perspective on Expert Routing in Bilingual MoE Language Models 2026

ARR Submission | arXiv Preprint | First Author

- Asked whether expert routing in bilingual English–German MoE language models primarily reflects language identity and lexical information or higher-level grammatical and syntactic structure.
- Designed controlled bilingual corpora, curriculum and no-curriculum training schedules, and held-out lexical, grammatical, and syntactic probes to separate these explanations.
- Analyzed routing with entropy, mutual and conditional mutual information, Jensen–Shannon divergence, frequency controls, and targeted routing ablations.
- Led the research question, experimental design, end-to-end analysis pipeline, and manuscript preparation.

Having a State Is Not Knowing It 2026

Project | Report | Apart Research Sprint | Co-author

- Investigated whether language models can report causally relevant internal decision states, separating possession of a state from verbal access to it through a falsifiable hierarchy of introspection capabilities.
- Replicated concept-injection experiments, then built a controlled Qwen benchmark using measured logit-margin shifts as ground truth for self-report, external decoding, and counterfactual intervention prediction.
- Native self-report reached **0.24 F1**, trainable self-report **0.52 F1**, and external probes decoded the state perfectly. Models predicted intervention direction at **0.70 F1** but not magnitude, separating qualitative state sensitivity from quantitative self-modeling.

Research Experience

NLP Research Intern — 2026

National Institute of Technology Tiruchirappalli | Submitted to ICON 2026 | Paper | Code

- Under faculty mentorship, investigated computational modeling of Classical Tamil *urai*, focusing on what linguistic structure and learned representations can recover in a low-resource setting.
- Developed mBART-style encoder–decoder and decoder-only generation/evaluation pipelines, alongside syntax-aware tags and explicit linguistic features as candidate inductive biases.
- Tested out-of-domain generalization on Thirukural couplets and analyzed clause structure, predicates, finite and non-finite constructions, and morphosyntactic signals for retrieval and representation learning.

Supporting Multimodal and Benchmark Work

Exploring Swin Transformers, Luminance Input, and Hybrid Architectures for Synthetic Image Detection — 2026

MediaEval 2026 Working Notes | Paper | Code | First Author

- Compared Swin, luminance-only, CNN–sequence, and hybrid architectures to test whether luminance and localized patch evidence improve robustness across synthetic-image generators; the strongest Swin system achieved **0.6997 F1**.

FinMMEval 2026 Shared Task Submissions – TextSentinels — 2026

CLEF 2026 Working Notes – Accepted | Papers: T1, T2, T3 | Co-author

- Built multilingual retrieval, grounded financial QA, and decision-making systems spanning FAISS model routing, BM25 over SEC filings and tables, and sentiment-aware trading; ranked **6th/6th/7th** in English/Chinese/Arabic MCQA.
- Improved local financial-QA ROUGE-1 F1 from **0.04 to 0.28**; the forward-tested trading system reached **1.14 cumulative return** and **2.03 Sharpe ratio**.

Technical Skills

ML / NLP	PyTorch, Hugging Face Transformers, scikit-learn, FAISS, Stanza
Analysis Methods	Mutual and conditional mutual information, routing entropy, Jensen–Shannon divergence, permutation tests, CCA, concept injection, external probes
Programming / Tools	Python, C, Java, TypeScript, Git, Linux, LaTeX

Academic Standing and Service

- **Merit Scholarship**: 3rd Position, First Year.
- **Reviewer / Sub-reviewer, CLEF 2026 Working Notes**: reviewed methodological clarity, experimental validity, presentation, and reproducibility.